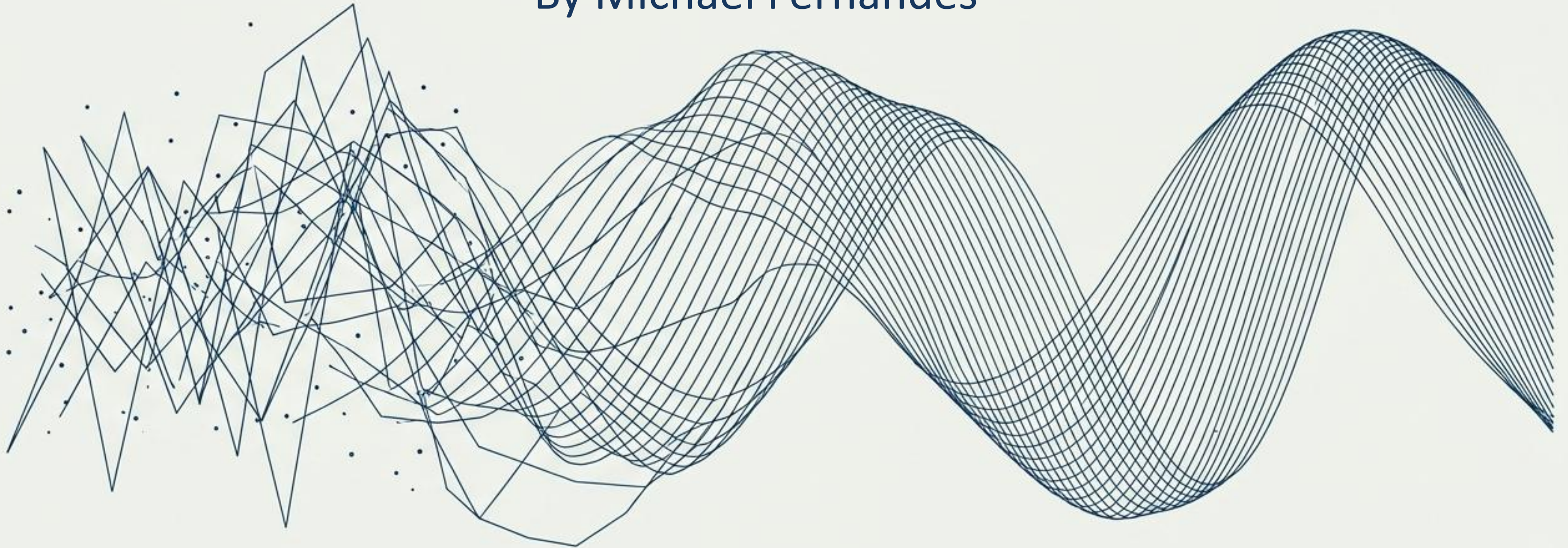


Types Of Quantum Monte Carlo Simulations

And Advanced Architectures of Stochastic Simulation in Quantitative Finance

By Michael Fernandes

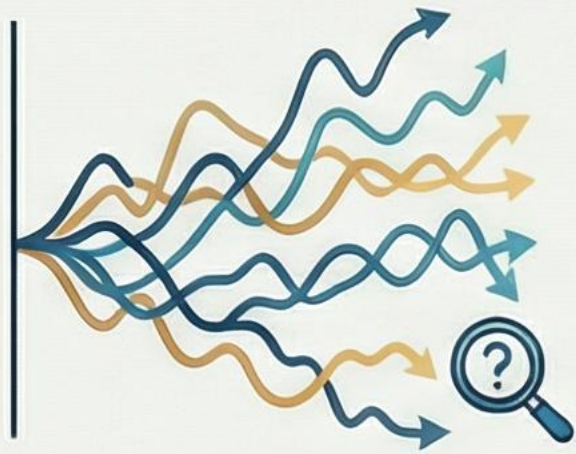


Project Link - <https://montecarlolab.streamlit.app/>

Smarter Simulations: The Quant's Guide to Advanced Monte Carlo Methods

Standard Monte Carlo simulations are slow for modern quantitative demands. Advanced methods accelerate convergence, reduce variance, and solve complex pricing problems.

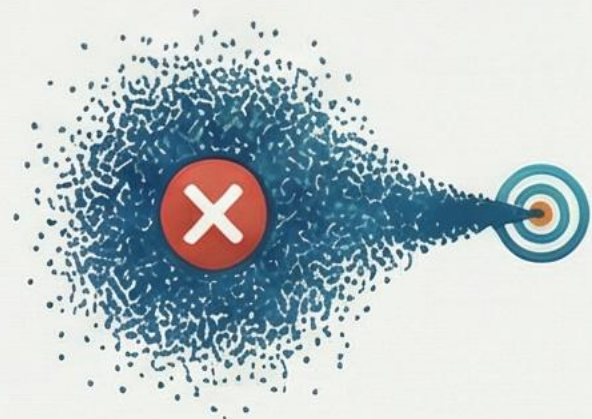
The Foundation & The Flaw



What is Monte Carlo Simulation?

Solves complex pricing problems by averaging thousands of simulated future asset paths.

The Core Limitation: Slow Convergence



10x ↑ Precision ↓ 100x Computations

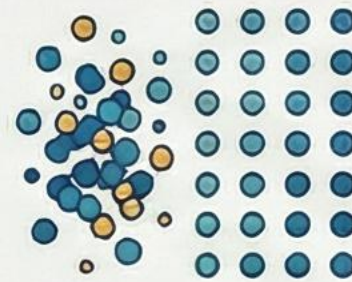
To improve precision by a factor of 10, you must increase computations by 100x.

$\Delta \rightarrow \Gamma$

The Challenge of 'Greeks'

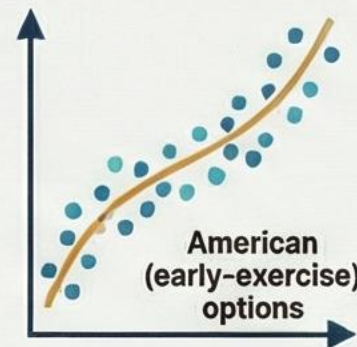
Calculating risk sensitivities like Delta and Gamma is computationally expensive and prone to noise.

The Modern Quant's Toolkit



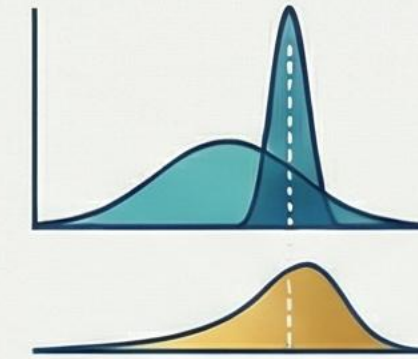
Quasi-Monte Carlo (QMC)

Replaces random numbers with uniform sequences, providing a significant speedup.



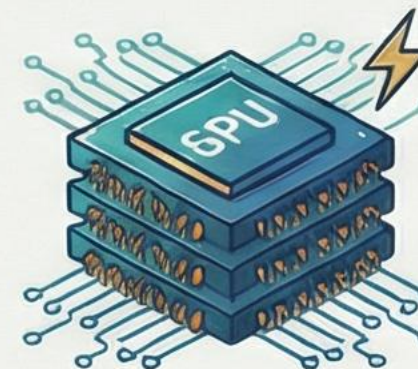
Least Squares Monte Carlo (LSMC)

A breakthrough using regression to solve the complex pricing of American (early-exercise) options.



Variance Reduction Techniques

Statistical methods like Antithetic and Control Variates squeeze more precision from fewer simulations.



High-Performance Computing (GPU)

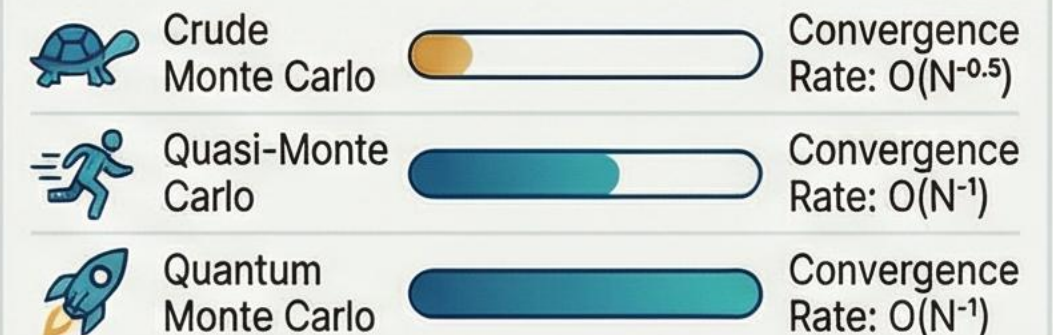
The parallel nature of simulations enables massive (500x+) speedups on GPUs.



The Quantum Future

Quantum algorithms promise an exponential speedup, fundamentally changing the economics of risk management.

Efficiency Gains



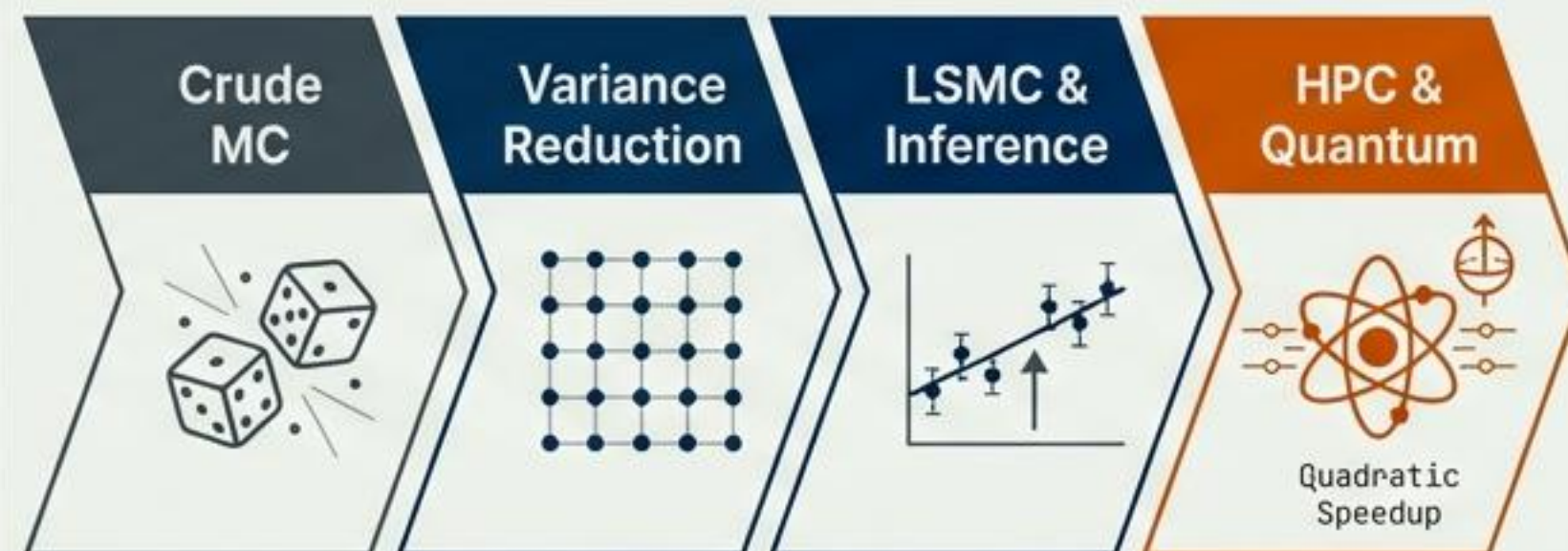
Executive Summary: The Evolution of Valuation Engines

Left Column (40%)

The transition from “Vanilla” to “Exotic” financial instruments has rendered closed-form analytical solutions insufficient. Monte Carlo (MC) simulation has emerged as the primary engine for valuation, risk management, and calibration. This deck analyzes the architectural evolution of stochastic methods:

- **From Random to Quasi-Random:** Overcoming the $O(N^{-1/2})$ convergence barrier using Low-Discrepancy Sequences.
- **Algorithmic Breakthroughs:** Solving the “American Option” optimal stopping problem via Least Squares Monte Carlo (LSMC).
- **Inference & Calibration:** Utilizing Markov Chain Monte Carlo (MCMC) and Particle Filters (SMC) for parameter estimation.
- **The Computational Frontier:** Leveraging GPU parallelism and the emerging quadratic speedup of **Quantum Amplitude Estimation**.

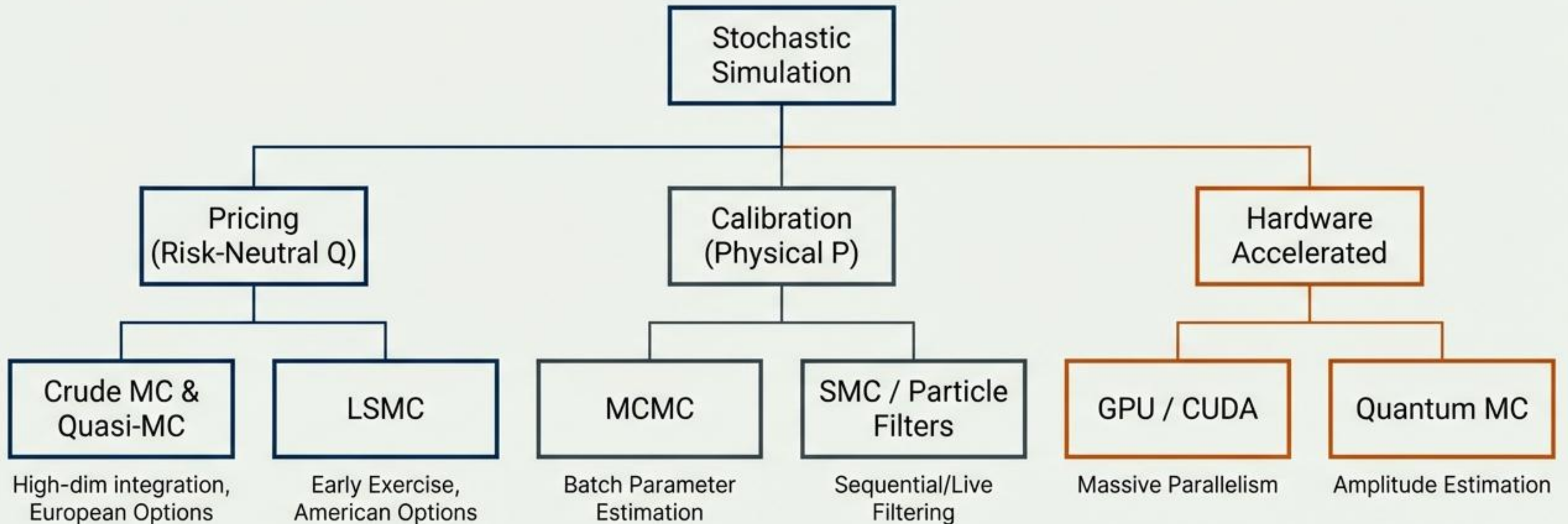
Right Column (60%)



Key Insight: Modern quantitative finance is no longer about accepting randomness, but **engineering it for maximum computational efficiency**.

Types Of Monte Carlo Simulations

A Taxonomy of Stochastic Architectures



Theoretical Foundations

The Risk-Neutral Framework & Discretization

The Fundamental Theorem of Asset Pricing states that derivative prices are expectations of discounted payoffs under the risk-neutral measure (Q).

$$V_t = \mathbb{E}^Q \left[e^{-\int (r_u du)} H(S_T) \mid \mathcal{F}_t \right]$$

The Simulation Mechanism:

Since analytical integration is impossible for complex derivatives, we invoke the Law of Large Numbers (LLN). We simulate N paths of the underlying asset and compute the arithmetic mean of the payoffs.

The Discretization Bias:

Assets follow continuous SDEs, but simulations are discrete (Euler-Maruyama Scheme).

$$X_{t+dt} = X_t + \mu(X_t, t)dt + \sigma(X_t, t)\sqrt{dt}Z$$

This introduces “Weak Error” distinct from statistical noise, critical for path-dependent options.

Discretization Bias



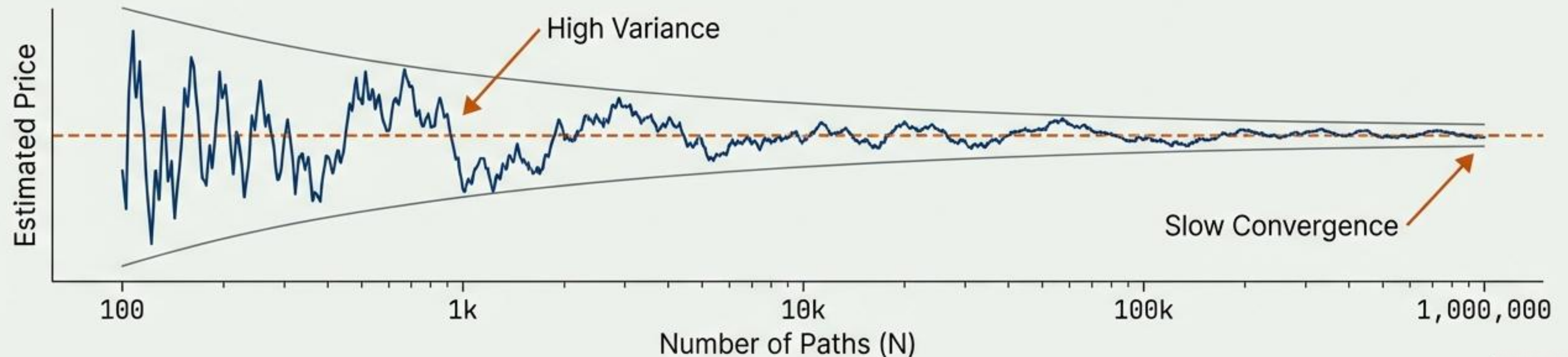
The Convergence Challenge

The Curse of $O(N^{-1/2})$

The defining characteristic of crude Monte Carlo is its slow convergence rate, governed by the Central Limit Theorem (CLT). The statistical error is standard deviation scaled by $1/\sqrt{N}$.

The Square Root Rule: To reduce error by a **factor of 10 (one decimal place)**, **computational effort (N) must increase by a factor of 100**.

Convergence Plot



Quasi-Monte Carlo (QMC)

Engineering Deterministic Uniformity

QMC replaces pseudo-random numbers with deterministic **Low-Discrepancy Sequences** (LDS) like **Sobol**, **Halton**, or **Faure**.

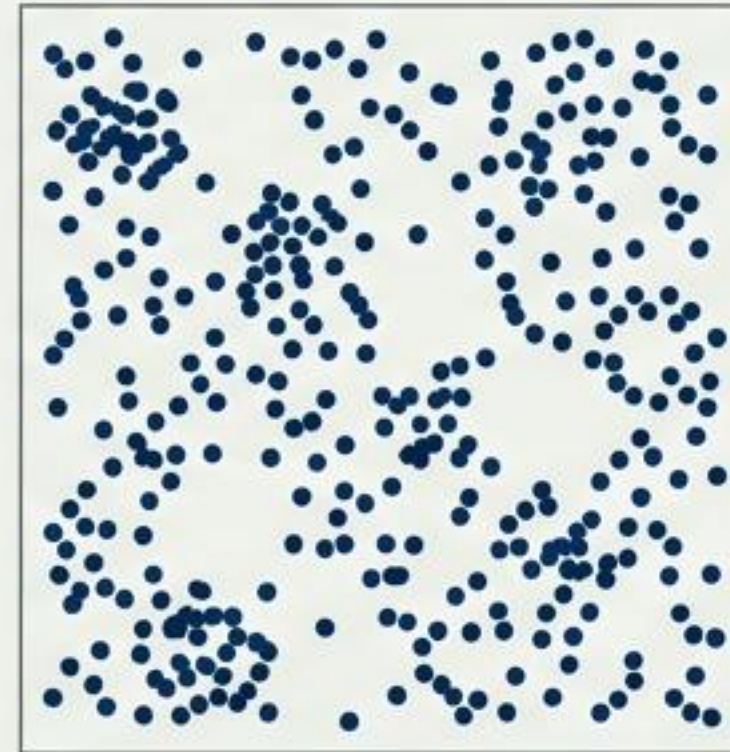
The Mechanism:

These sequences are designed to be **self-avoiding** and **space-filling**, minimizing “**discrepancy**”—the deviation from perfect uniformity.

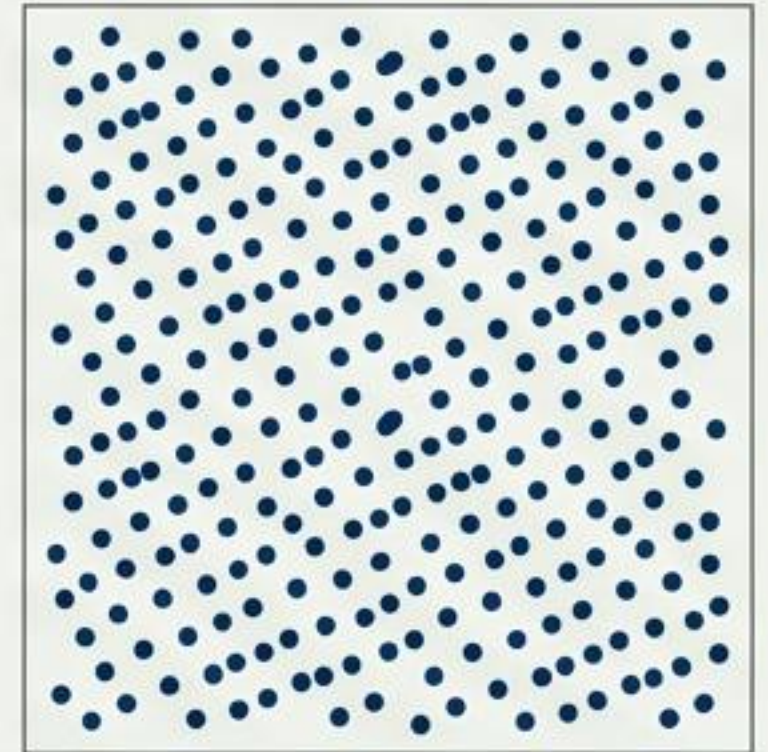
Performance Gains:

- Convergence approaches $O(1/N)$, offering a quadratic speedup over standard MC.
- Theoretical basis: The **Koksma-Hlawka Inequality**.

A. Pseudo-Random
(Standard MC)



B. Low-Discrepancy
(Sobol)



Engineered Uniformity

Effective Dimension & Brownian Bridges

Navigating High-Dimensional Space: A daily monitored option implies 252 dimensions. Why does QMC not degrade? Because financial problems have low '**Effective Dimension**'. **Variance** is driven by the first few principal components (market trends), not daily noise.

The **Brownian Bridge Technique**:

We restructure path generation to match the Sobol sequence properties:

1. Generate the **End Point (T)** first (using the most uniform dimensions).
2. Generate the **Mid Point (T/2)**.
3. Fill in **quarters**, then **eighths** recursively.

Brownian Bridge Construction



Variance Reduction I: Antithetic Variates

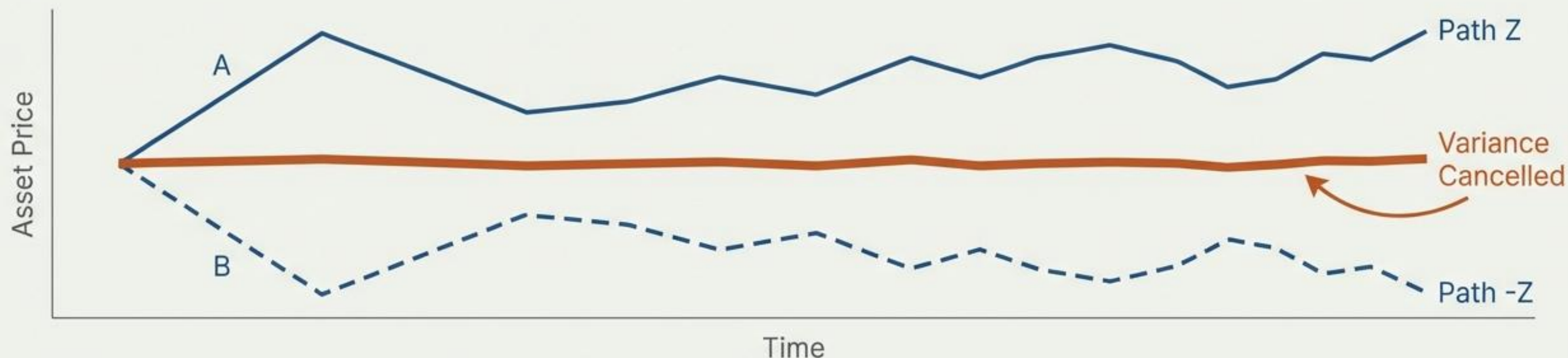
Exploiting Symmetry for Stability

Antithetic Variates (AV) is the most robust variance reduction technique.

Mechanism: For every path driven by random draws $\mathbf{Z}$, generate a companion path using $-\mathbf{Z}$.

$$V_{AV} = \frac{Y(\mathbf{Z}) + Y(-\mathbf{Z}))}{2}$$

Because payoffs are negatively correlated, averaging them cancels out extreme deviations.



Variance Reduction II: Control Variates

Leveraging Analytical Proxies. Often the most powerful technique, reducing variance by orders of magnitude.

The Logic: To price a complex 'Exotic' option (e.g., Arithmetic Asian), we simulate a 'Control' instrument (e.g., Geometric Asian) that is highly correlated but has a known closed-form solution.

Estimator:

$$V_{cv} = V_{\text{exotic}} - \beta (V_{\text{control}} - \text{Price}_{\text{Analytical}})$$

We subtract the *error* of the control simulation from the exotic simulation to clean the noise.



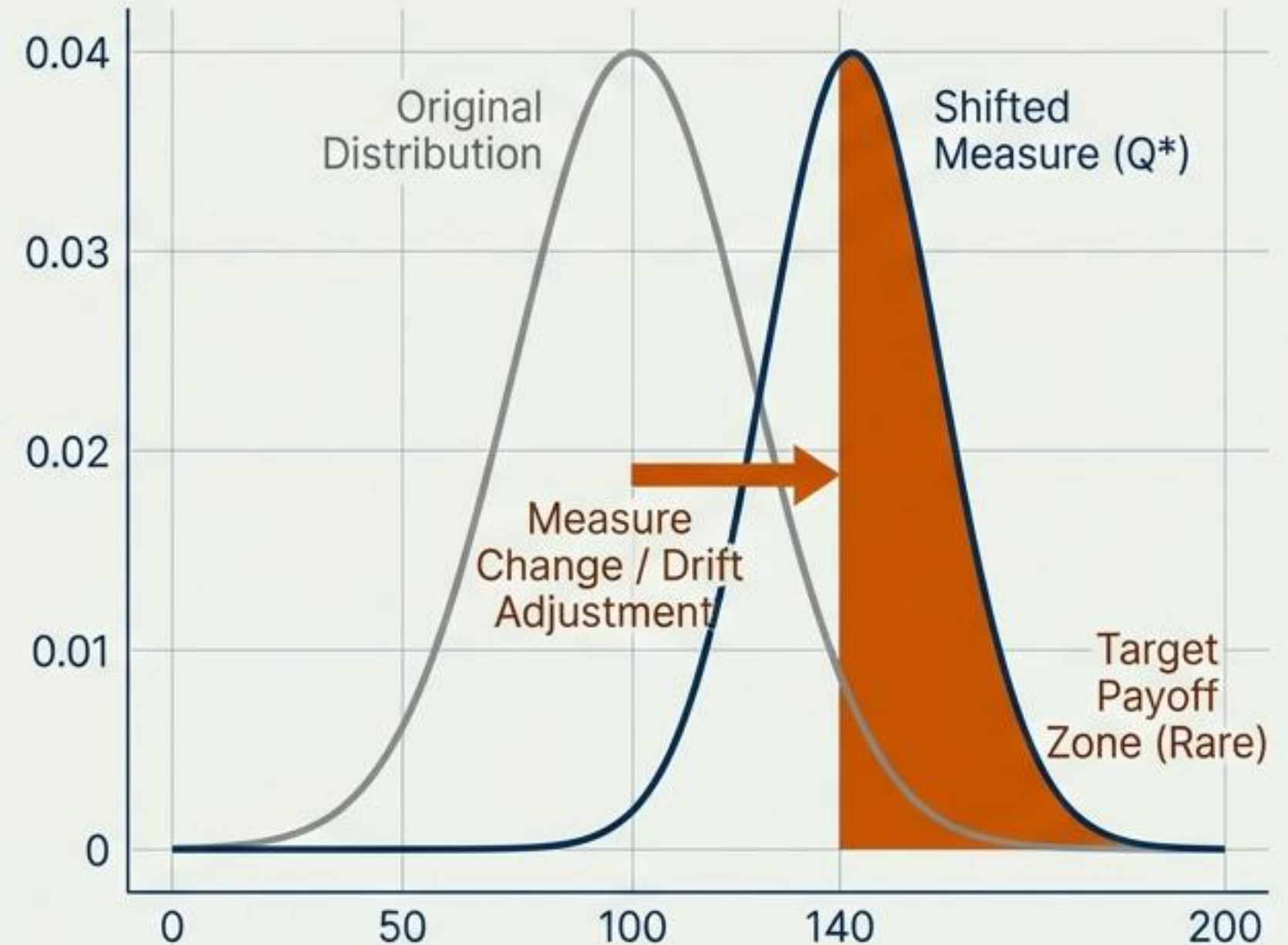
Variance Reduction III: Importance Sampling

Simulating the Improbable

Standard MC is wasteful for Deep Out-of-the-Money (OTM) options, where 99.9% of paths result in zero payoff.

The Mechanism: Change the probability measure to force the simulation to sample “important” regions (e.g., increasing drift so the asset ends In-The-Money).

Correction: Payoffs must be weighted by the Radon-Nikodym derivative (Likelihood Ratio) to correct the bias.

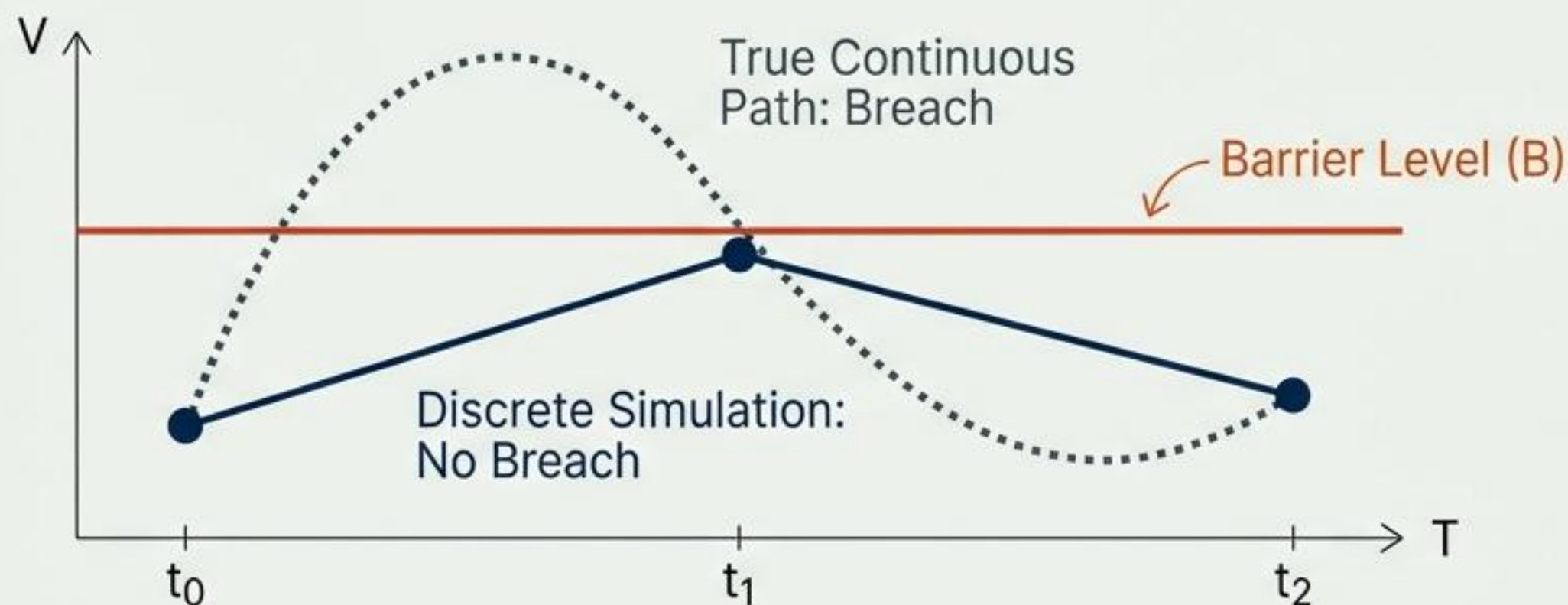


Pricing Path-Dependent Derivatives

Path-dependent options require simulating the entire trajectory.

Discrete simulations create specific risks.

The Hidden Barrier Breach



Continuity Correction: We must calculate the probability of the bridge maximum exceeding B , even if end-points are safe.

The American Option Challenge

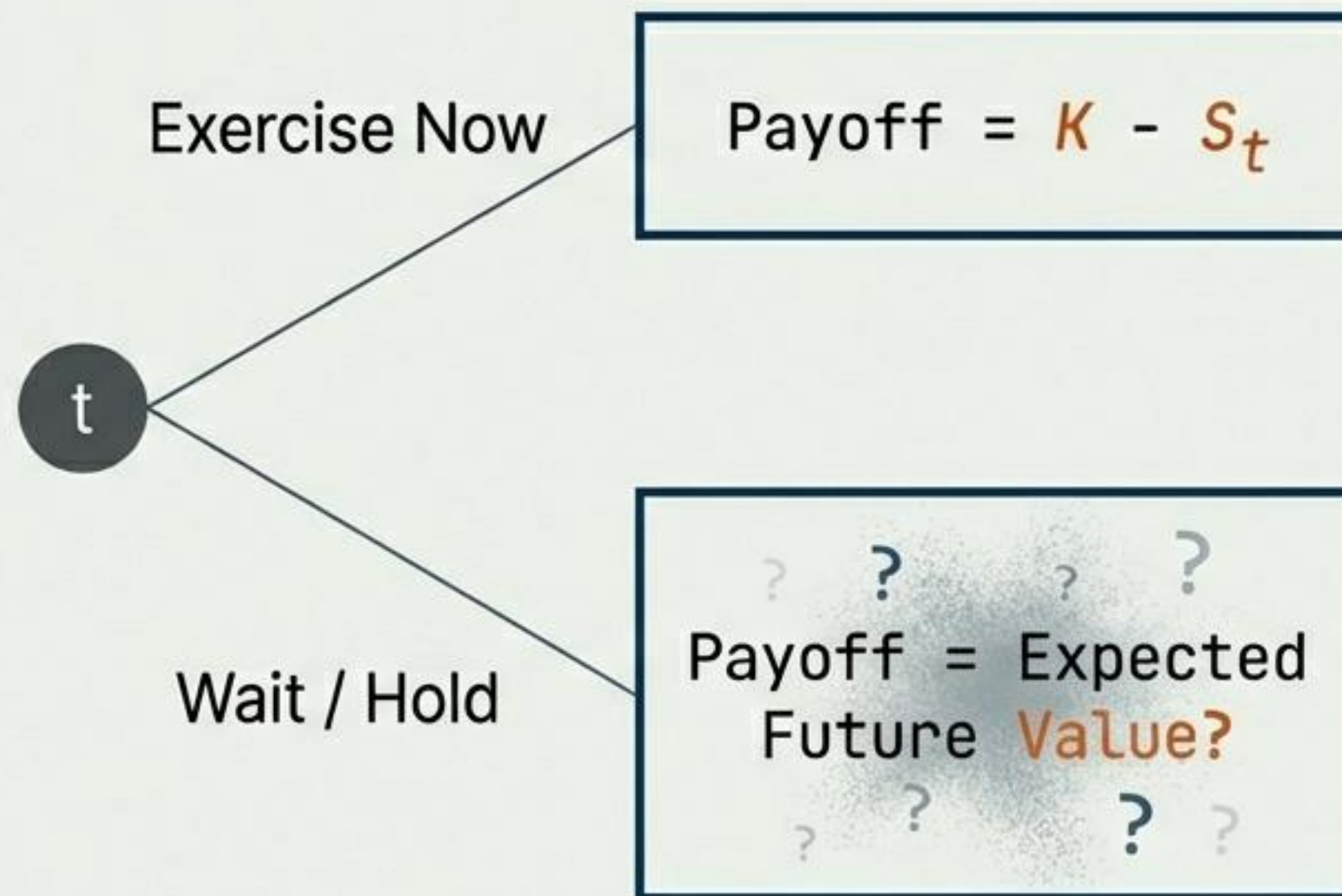
The Optimal Stopping Problem

American options allow exercise at any time. This creates a recursive optimization problem.

Decision Rule: Exercise if Intrinsic Value > Continuation Value (Expected value of holding).

The Simulation Paradox:

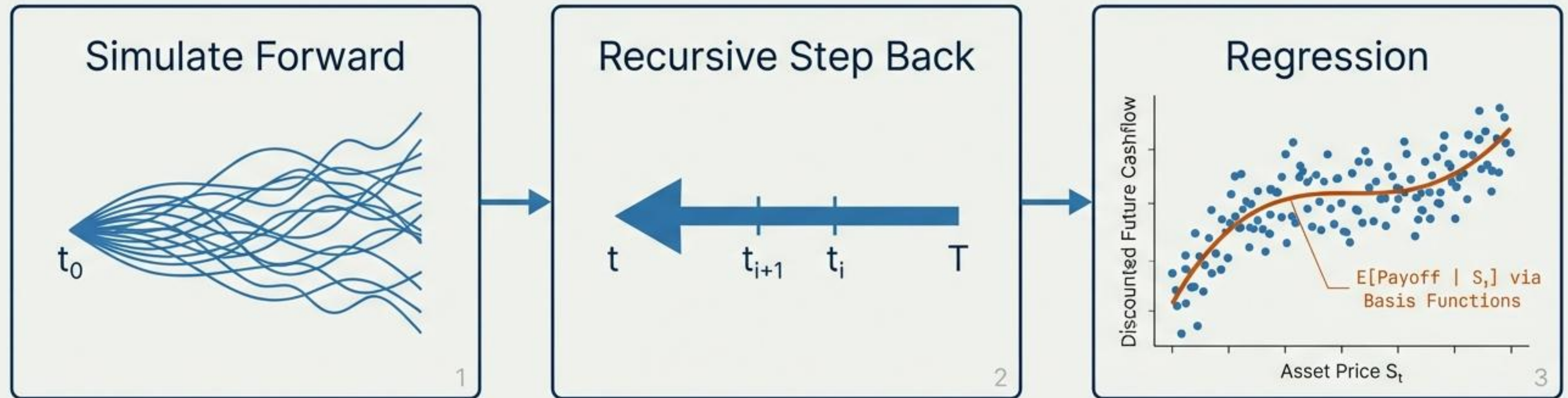
- Standard MC moves forward in time.
- Continuation Value depends on the future, which is unknown at time t .
- Binomial Trees work backward but suffer the 'Curse of Dimensionality' in multi-asset problems.



Least Squares Monte Carlo (LSMC)

The Longstaff-Schwartz Algorithm (2001)

LSMC revolutionized pricing by combining simulation with regression to estimate continuation values.



Advantage: Complexity grows linearly with dimension, enabling high-dimensional American Basket pricing.

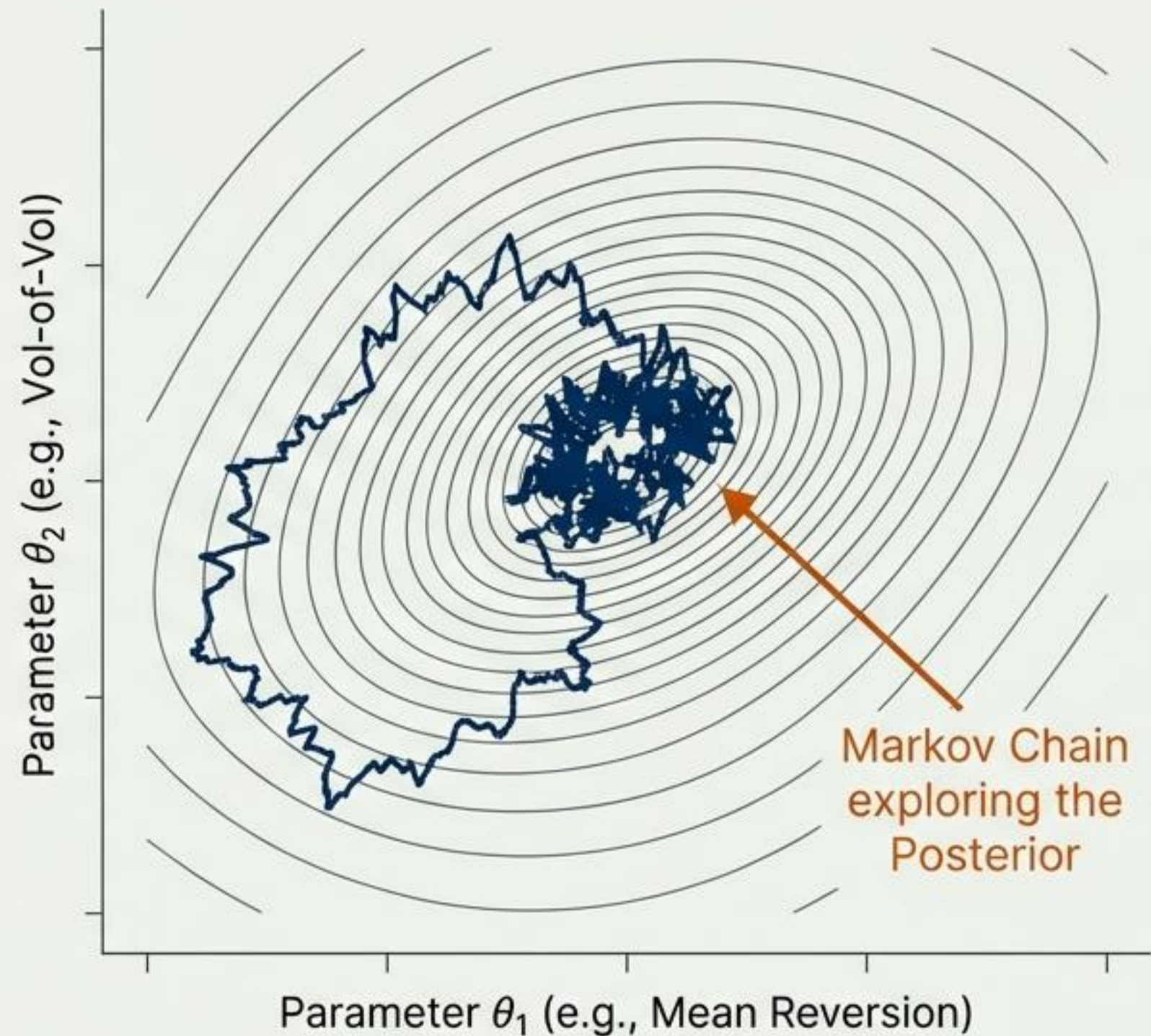
Model Calibration: MCMC

From Pricing (Q) to Inference (P).

Calibration involves finding parameters (e.g., Heston Vol-of-Vol) that fit market data.

Bayesian Inference (Metropolis-Hastings): We seek the posterior distribution of parameters.

1. Propose new parameter set.
2. Accept/Reject based on likelihood ratio.
3. Result: A chain of samples quantifying uncertainty.



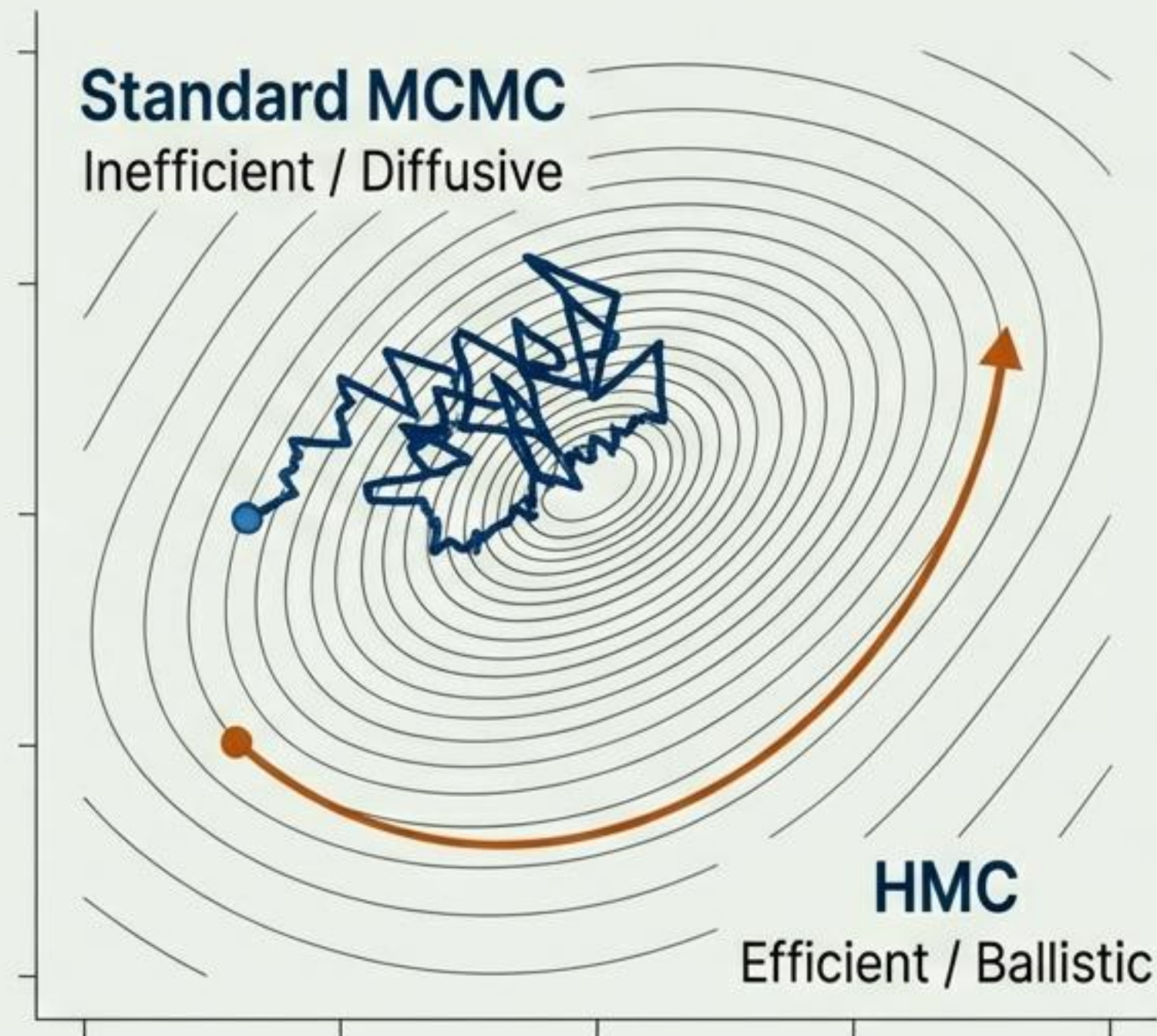
Advanced Calibration: Hamiltonian Monte Carlo (HMC)

Standard MCMC suffers from random walk behavior. HMC uses physics-informed gradients to propose efficient jumps.

The Mechanism:

Treat parameters as particles sliding on a frictionless surface defined by the negative log-posterior (potential energy).

- Momentum 'kick' starts the motion.
- Hamiltonian dynamics guide the flow toward high probability regions.
- Essential for calibrating 50+ parameter models.



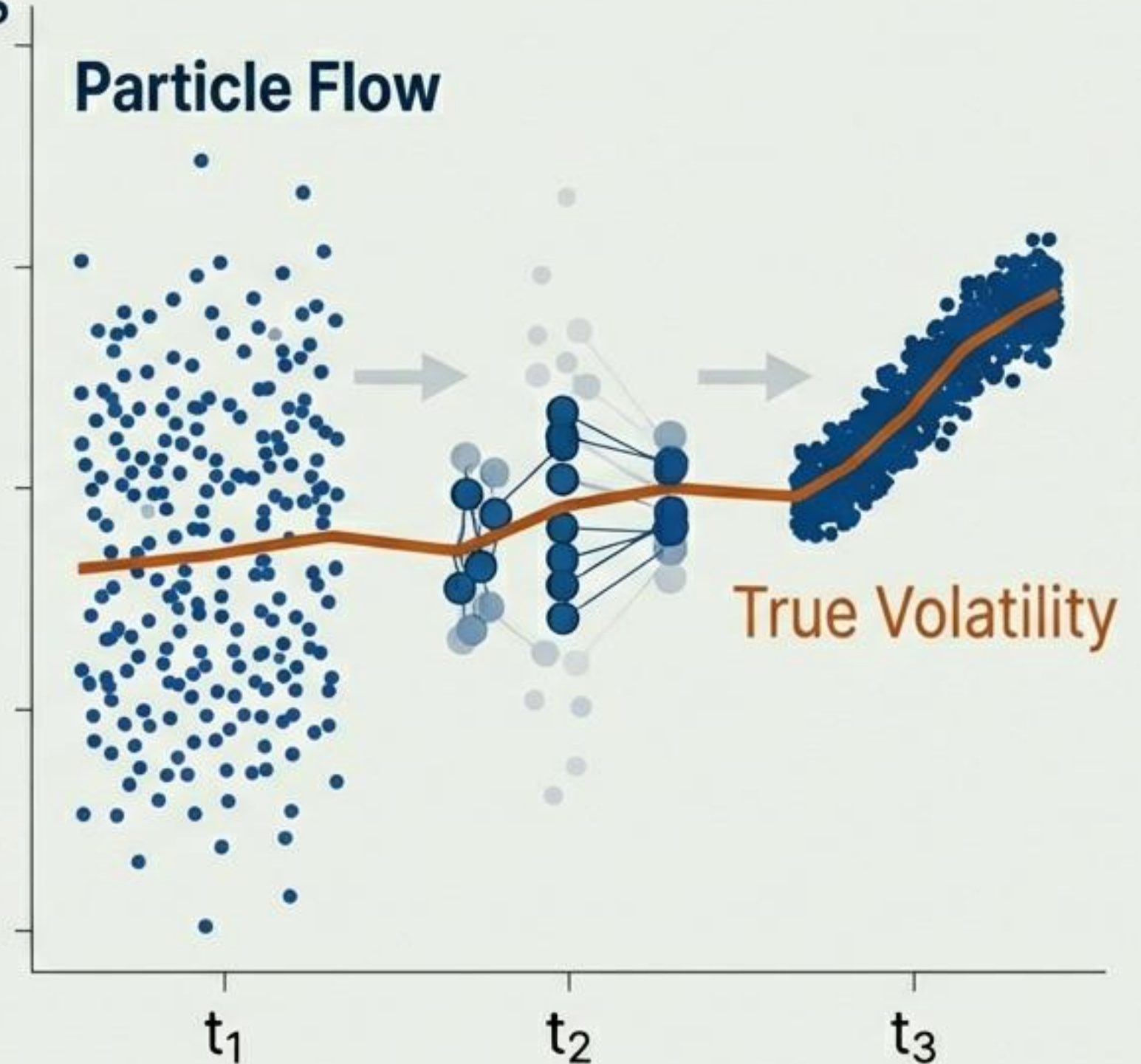
Latent State Filtering: SMC

Sequential Monte Carlo / Particle Filters

Unlike batch MCMC, **SMC** allows for **online, recursive** estimation of hidden states (e.g., Stochastic Volatility) as data arrives.

The Algorithm:

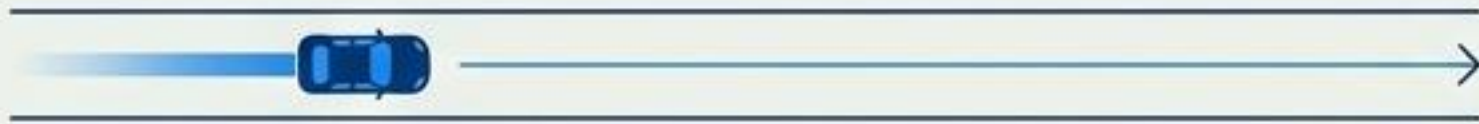
1. **Predict:** Propagate N particles forward.
2. **Update:** Reweight particles based on new market data likelihood.
3. **Resample:** Discard low-weight particles, duplicate high-weight ones to prevent degeneracy.



High-Performance Computing (HPC)

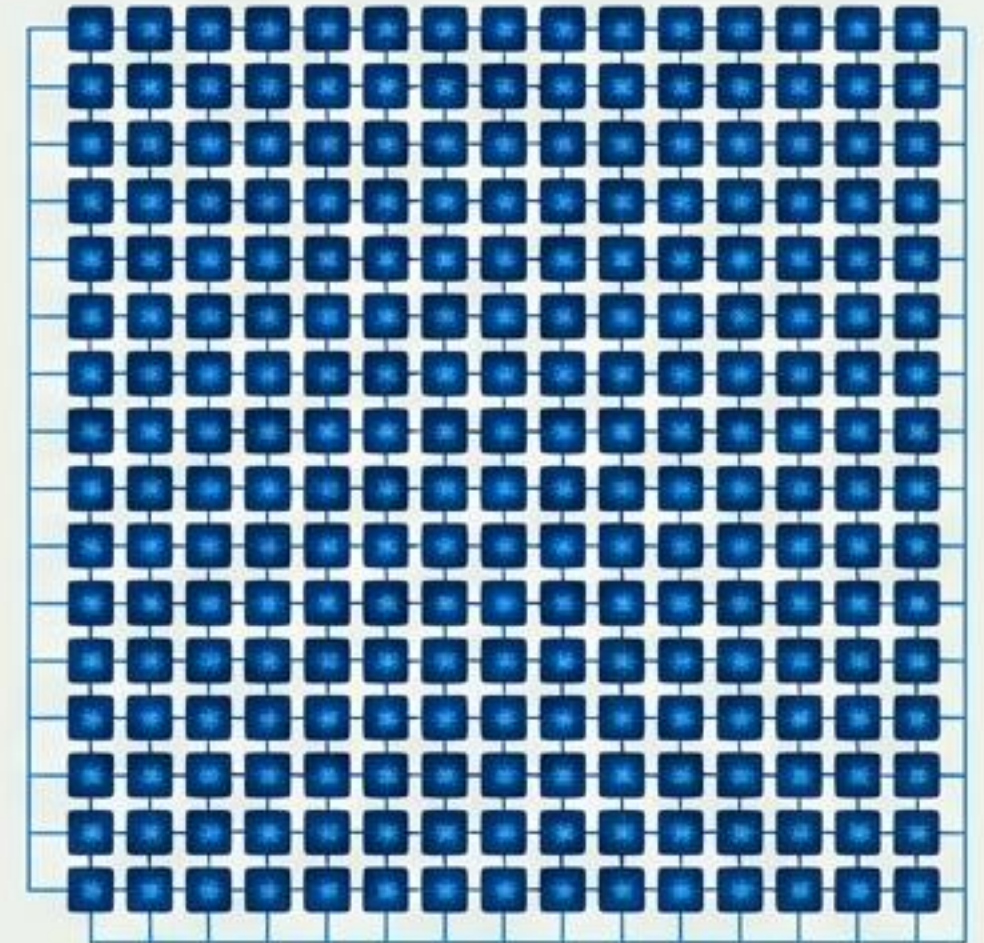
Monte Carlo is “Embarrassingly Parallel”. Each path is independent, making GPUs the ideal architecture.

CPU (Serial)



500x Speedup →
(Seconds -> Milliseconds)

GPU (CUDA SIMT)



Challenge: Standard RNGs fail in parallel. Requires Parallel RNGs (MTGP32) to ensure independent streams. Distributed Spark clusters are used for bank-wide XVA calculations.

The Future: Quantum Monte Carlo (QMC)

The Quadratic Speedup via Amplitude Estimation

As silicon nears limits, Quantum Computing offers a paradigm shift.

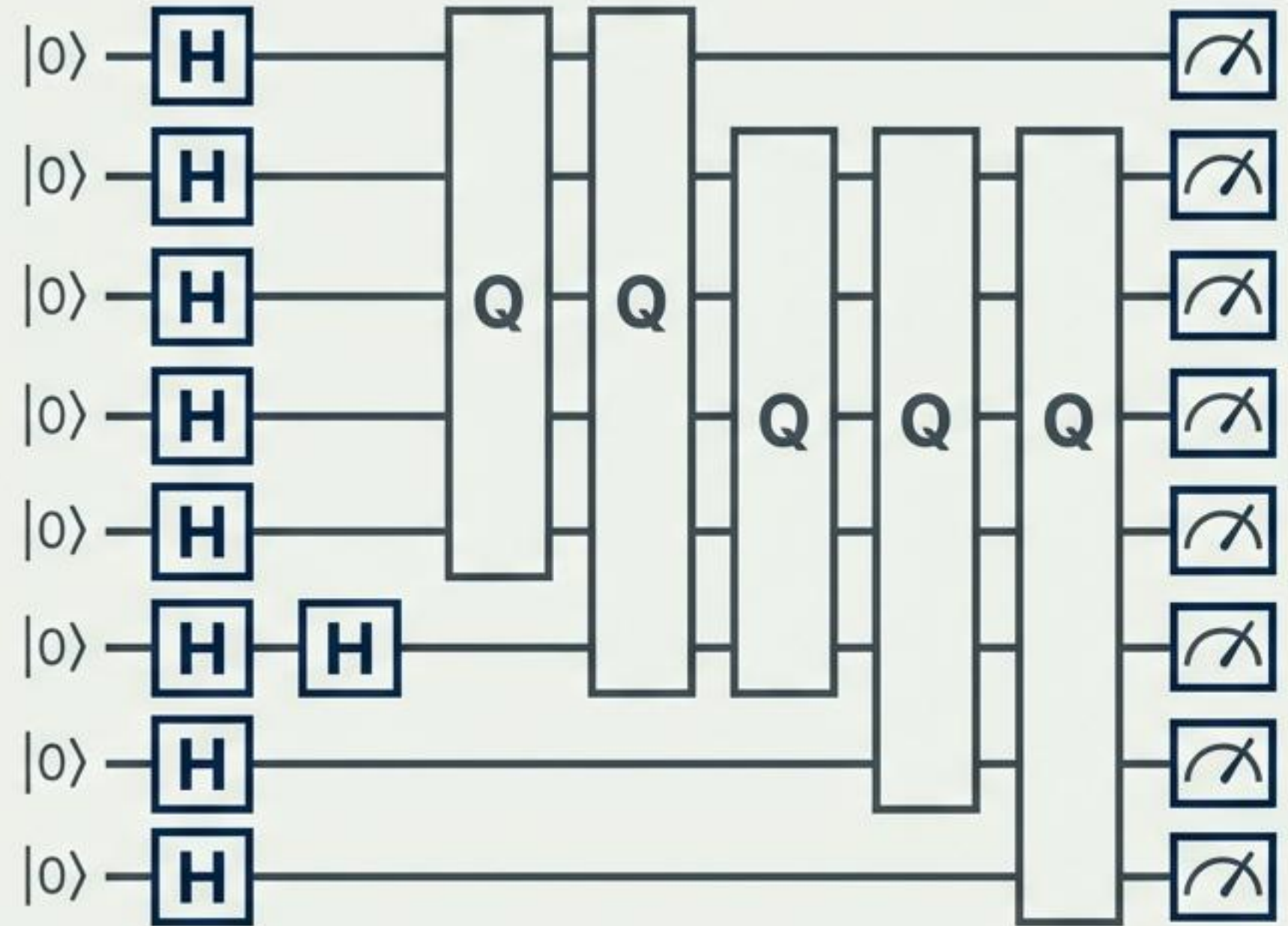
Note: Here QMC refers to Quantum algorithms, not Quasi-Monte Carlo.

Quantum Amplitude Estimation (QAE):

- Encodes probability into quantum state amplitudes.
- Convergence scales as $O(M^{-1})$ (Inverse) vs $O(N^{-1/2})$ (Inverse Sq Root).

To gain 10x precision:

- Classical: 100x ops
- Quantum: 10x ops



$$|\psi\rangle = \sum(\sqrt{p_i})|i\rangle$$

Strategic Synthesis & Comparison

Method	Best Application	Key Advantage	Convergence
Quasi-MC	High-dim European / Path-dependent	Deterministic Uniformity	$\sim O(N^{-1})$
Antithetic/Control	Vanilla / Asian Options	Variance Reduction	Factor Reduction
LSMC	American / Bermudan	Early Exercise Handling	Regression-based
HMC / MCMC	Model Calibration	Posterior Exploration	Sample Efficiency
Quantum MC	Future Ultra-High Precision	Quadratic Speedup	$O(N^{-1})$

Conclusion: The future belongs to hybrid architectures—combining the mathematical elegance of Variance Reduction with the brute force of GPUs and the emerging power of Quantum algorithms.